

AIRWISE

India Air Quality Intelligence Dashboard

Professional Project Report
Streamlit • Pandas • NumPy • Plotly

An interactive data-analysis and visualization project built from the uploaded India air-quality CSV dataset. All dataset-specific figures in this report are calculated from the supplied file.

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1. Introduction

AIRWISE – India Air Quality Intelligence Dashboard is a Streamlit-based analytics application designed to make air-quality observations easier to explore at national, state, city, station and pollutant levels. The dashboard combines data cleaning, descriptive analysis, interactive visualization, geographic exploration and automatically generated data-driven insights.

The supplied application code implements five main sections: National Overview, India Map, Pollutant Analysis, State & City Comparison, and AI/Data-Driven Insights. The project README describes the same structure and identifies the dataset fields used by the application. ■filecite■turn0file0■L60-L95■ ■filecite■turn0file1■L10-L16■

2. Objectives

- Inspect and prepare the supplied air-quality dataset.
- Provide a clear overview of pollutant levels and temporal patterns.
- Enable geographic exploration using station latitude and longitude.
- Compare pollutant levels across states and cities.
- Generate insights directly from the observed dataset rather than using unsupported or fabricated predictions.
- Package the analysis as a professional portfolio-ready Streamlit project.

3. Dataset

The uploaded file is **Air quality in india.csv**. The dataset contains **3,514** rows and **11** columns in its raw form. The project README identifies the principal fields as country, state, city, station, last_update, latitude, longitude, pollutant_id, pollutant_min, pollutant_max and pollutant_avg. ■filecite■turn0file1■L20-L26■

Metric	Observed value
Raw rows	3,514
Raw columns	11
States represented	30
Cities represented	261
Stations represented	502
Pollutant types	7
Pollutants	CO, NH3, NO2, OZONE, PM10, PM2.5, SO2
Observed start	2026-09-22 22:00:00
Observed end	2026-09-22 22:00:00

4. Data Cleaning & Quality Checks

Missing values: the raw CSV contains 990 missing cells across all columns. The supplied app performs median imputation for the pollutant_min, pollutant_max and pollutant_avg fields when missing values are present.

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Duplicate check: 0 completely duplicated rows were identified in the raw dataset. The supplied application code does not currently drop duplicates, so this result is reported as a quality finding rather than silently altering the application logic.

Date correction: the dashboard parses last_update using the DD-MM-YYYY HH:MM:SS format with invalid values coerced to missing; 0 rows had an unparseable/missing timestamp and were excluded by the application's date-cleaning step. ■filecite■turn0file0■L27-L35■

Numeric pollutant fields: pollutant_min, pollutant_max and pollutant_avg were treated as numeric measures for the report's descriptive analysis. The dashboard uses pollutant_avg for most comparisons and pollutant_min/pollutant_max when examining ranges.

Missing-value profile

Column	Missing cells
pollutant_min	330
pollutant_max	330
pollutant_avg	330

5. Methodology

The analytical workflow follows a practical exploratory-data-analysis pipeline: load the CSV, inspect its structure, correct the timestamp field, handle missing pollutant values, derive a date field, aggregate observations by geography/pollutant/time, and present the results through interactive Streamlit components.

The dashboard uses Streamlit for the application layer, Pandas for tabular processing, NumPy for numerical operations, and Plotly Express for interactive charts and maps. These technologies match the project's README and requirements. [filecite turn0file1 L28-L34](#) [filecite turn0file2 L2-L5](#)

6. Dashboard Features

National Overview: KPI-style summaries and a time-series view of average pollutant levels. The application calculates overall averages and identifies states/cities with extreme average values within the active filters.

[filecite turn0file0 L60-L90](#)

India Map: Plotly Mapbox visualization using latitude and longitude. Markers are aggregated by state, city and coordinates and sized/colored using average pollutant levels. [filecite turn0file0 L95-L120](#)

Pollutant Analysis: state-level pollutant comparison and distribution analysis using interactive bar charts and box plots. [filecite turn0file0 L122-L139](#)

State & City Comparison: ranked city and state comparisons for the selected pollutant, including highest and lowest average city groups. [filecite turn0file0 L141-L173](#)

AI/Data-Driven Insights: deterministic insights derived from the current filtered data, including observed temporal direction, highest state-level average, highest overall pollutant average and widest observed city range. These are analytical summaries, not machine-learning forecasts. [filecite turn0file0 L175-L215](#)

7. Exploratory Data Analysis & Findings

The following findings are descriptive statistics calculated directly from the supplied CSV after the report's cleaning steps. They should be interpreted as characteristics of this dataset and observation period, not as a complete assessment of India's air quality.

Pollutant-level descriptive statistics

Pollutant	Observations	Mean	Median	Min	Max
PM10	502	76.86	63.50	12.00	342.00
PM2.5	502	68.26	49.50	2.00	405.00
CO	502	37.26	31.00	2.00	195.00
OZONE	502	31.50	26.00	0.00	268.00
NO2	502	23.51	21.00	0.00	163.00
SO2	502	14.58	12.00	0.00	108.00
NH3	502	8.53	5.00	0.00	31.00

Across all pollutant observations combined, the state-level mean of pollutant_avg is highest for **Ladakh** (81.14) and lowest for **Sikkim** (6.14) in this dataset. Because different pollutants may be measured on different scales and units, this combined comparison should not be interpreted as a universal air-quality ranking.

At city level, the highest combined mean pollutant_avg is observed for **Khora** (101.86), while the lowest is **Gangtok** (6.14). These values are descriptive and reflect the pollutant mix represented in the file.

The cleaned timestamp coverage extends from **2026-09-22 22:00:00** to **2026-09-22 22:00:00**. The dashboard derives a calendar date from last_update to support time-series aggregation.

8. Technical Implementation

The Streamlit application is intentionally contained in a single app.py file. Data loading is cached with Streamlit's `cache_data` mechanism. Global state and city filters are applied before the analytical sections, while pollutant selection controls the map, comparisons and insights. [filecite turn0file0 L12-L40](#)

Plotly is used for the line chart, state comparison bars, distribution box plots, city rankings and geographic map. This provides hoverable and filter-responsive visual exploration without introducing a separate frontend framework.

9. Limitations

- The dataset represents the observations present in the supplied CSV; it does not necessarily cover every location, station or time period in India.
- Pollutant values may have different measurement units and scales, so cross-pollutant numerical comparisons require care.
- Median imputation is used for missing pollutant measures in the supplied application and may reduce the apparent variability of those fields.
- The current application checks data quality but does not explicitly remove duplicate rows before analysis.
- The AI/Data-Driven Insights section uses deterministic descriptive rules; it is not a generative AI model and does not provide unsupported forecasts.
- Correlation, causal inference and health-risk assessment are outside the scope of this dashboard.

10. Future Scope

Potential future enhancements include a more formal data-quality audit panel, pollutant-specific units and thresholds, richer time-window controls, station-level drill-downs, downloadable filtered datasets, more advanced anomaly detection, and validated forecasting models where the dataset provides sufficient longitudinal coverage. Any future predictive layer should be evaluated separately and clearly distinguished from descriptive analysis.

11. Conclusion

AIRWISE provides a portfolio-oriented framework for exploring India's air-quality observations through an interactive Streamlit interface. Its value comes from connecting data preparation with intuitive geographic, temporal and comparative views while keeping the generated insights tied to the underlying dataset. The supplied README confirms the intended dashboard structure and local execution workflow. [filecite turn0file1 L36-L49](#)